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Research interests: Machine Learning systems, Model Deployment and Acceleration, Heterogeneous Computing, and Applications in High-Energy Physics.

Education

- 2023–2026 **University of Chinese Academy of Sciences (UCAS), Beijing**
M.Sc. in Computer Science
Supervisor: Prof. Cheng Yaodong (IHEP)
Thesis: AutoDeploy: An Intelligent Toolkit for ML Deployment on Heterogeneous Hardware in High-Energy Physics
- 2019–2023 **Yangzhou University, Jiangsu**
B.Sc. in Microelectronics Engineering
Thesis: A Smart Energy-Efficient and Cost-Effective Home Automation System for Developing Countries

Experience

- 2023–2026 **Institute of High Energy Physics (IHEP), Beijing** *Research Fellow*
 - LHAASO GRB sensitivity improvements via ML pipelines and Bayesian optimization
 - Hardware-aware *ParticleNet* deployment for CEPC and CMS (CERN)
 - *AutoDeploy*: AutoML toolkit for hardware- and goal-aware model deployment
- 2021–2022 **GeePower Energy Technology Co., Ltd., Yangzhou** *Web Dev*
 - Full-stack development of company website
 - Database design, queries, and optimization

Research Publications

Published

- 1 **A. A. Mustofa**, Y. Cheng, and Z. Yao, “Enhancing GRB detection: Bayesian optimization of triggerless data analysis for LHAASO-WCDA,” *EPJ Web of Conferences*, vol. 337, p. 01 330, 2025. [DOI](#): 10.1051/epjconf/202533701330.
- 2 Z. Minxing, F. Shiyuan, G. Yu, C. Yaodong, **A. A. Mustofa**, and C. Gang, “Fast lossless images compression for synchrotron radiation facility using Deep learning and Hybrid architecture,” *Radiation Detection Technology and Methods*, vol. 8(4), pp. 1693–1703, 2024. [DOI](#): 10.1007/s41605-024-00490-9.
- 3 **A. A. Mustofa**, Y. A. Dagnew, P. Gantela, and M. J. Idris, “SECHA: A smart energy-efficient and cost-effective home automation system for developing countries,” *Journal of Computer Networks and Communications*, vol. 1, p. 8 571 506, 2023. [DOI](#): 10.1155/2023/8571506.

Submitted

- 1 **A. A. Mustofa** and Y. Cheng, “A knowledge-driven framework for hep model deployment on heterogeneous hardware,” in *Radiation Detection Technology and Methods*, submitted, vol. 10, 2026.
- 2 L. Wei, Z. Meng, X. Li, **A. A. Mustofa**, Y. Cheng, and H. Li, “MCG: A multi-data-center content gateway for topology-hiding and secure wan storage access,” in *2026 IEEE International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, submitted, 2026.

- 3 M. Hasan, Y. Cai, A. T. Dufera, and **A. A. Mustofa**, “GL-VINS-Mono: Lidar-assisted monocular visual-inertial odometry in GNSS-denied environments,” in *IEEE Robotics and Automation Letters*, submitted, 2025.

Skills

Languages	English, Basic Mandarin, Amharic
Programming	Python, C++, Bash, PyTorch, TensorFlow, $\text{\LaTeX}$
Machine Learning	RL, MOBO, Graph ML (HGT, TGN), GFlowNet, Rainbow DQN, AutoML, RecSys & Policy Models, Deployment Optimization
Model Acceleration	Quantization, Pruning, HLS, TensorRT, ONNX Graph Optimization
Systems & Tooling	Docker, Linux, Git, CI/CD, Orchestration, Experiment Tracking
Hardware	NVIDIA GPUs, Xilinx FPGAs, Huawei Ascend NPUs, Microcontrollers, Embedded Systems
Databases / Storage	Neo4j, MySQL, EOS, SeaweedFS

Miscellaneous Experience

Awards and Achievements

- 2026 25th Director’s Grant Award for Best Graduation Projects.
- 2023-6 ANSO Scholarship for Young Talents.
- 2021 Yangjie “*Excellent Student*” Scholarship.

Certification

- 2022 Elements of AI. Awarded by Helsinki University.
- Technical Support Fundamentals. Awarded by Google.
- Programming Foundations with JavaScript, HTML and CSS. Awarded by Duke University.
- Cybersecurity for Everyone. Awarded by University of Maryland.
- HSK 4 Chinese Proficiency Certificate.

References

Available on Request